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IARCO Research Proposal

Confirmation Fatigue in AI-Agent Security: Balancing Human Oversight and Agent Autonomy

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Category	:	Junior
Research Topic	:	Confirmation Fatigue in AI-Agent Security
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International Academic Research Competition (IARCO)

Confirmation Fatigue in AI-Agent Security: Balancing Human Oversight and Agent Autonomy

Research Objectives:

1. To investigate how confirmation frequency affects users' ability to identify and appropriately respond to unsafe actions proposed by an AI agent.
2. To examine how confirmation frequency affects users' decision speed and confidence when evaluating AI-agent actions.
3. To examine how different confirmation frequencies may affect the balance between effective human oversight and AI-agent autonomy.

Research Question:

How does confirmation frequency affect users' accuracy, decision time, and confidence when identifying and responding to unsafe actions proposed by an AI agent?

Introduction:

AI agents are artificial intelligence systems that can independently plan and execute tasks for users by making decisions and interacting with digital tools and systems. Given a task, an AI agent can plan its actions, use external tools, evaluate its progress, and keep working toward completion. This ability allows AI agents to perform increasingly complex tasks with greater independence and efficiency. However, as AI agents gain greater autonomy, their actions can also have security and privacy implications, particularly when they interact with digital systems or handle sensitive information. Human supervision therefore continues to be an important safeguard, as it allows users to look over important actions before they take place. There is, however, another problem caused by frequent approval requests. Although the purpose of such prompts is to keep users in control, they may come to seem routine if seen too often, and as a result users might pay less attention and make quicker decisions without properly considering what the agent is about to do. This becomes especially important when a sensitive action appears among many ordinary ones. A user who has become used to approving routine requests may overlook a risky action or approve it without enough thought. In such cases, a safety feature meant to strengthen human oversight could actually become less effective and increase the risk to a user's privacy, data, or security.

Literature Review:

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Research on Automation Bias and Human Reliance on Automated Systems -

Automation can make decision-making easier, but relying on it too much can reduce how carefully people check the information themselves. Users may overlook important details that are not highlighted by the system or follow an incorrect recommendation even when other information suggests otherwise [8]. This tendency, known as automation bias, has been observed in different decision-making settings and can become stronger when tasks are complex or mentally demanding [4]. When checking an automated recommendation requires extra effort, users may be less likely to examine it carefully [6]. Attention therefore plays an important role in effective human oversight, especially when users are also dealing with other tasks at the same time [7].

Research on Alert Fatigue and Repeated Warnings -

Frequent or repeated alerts can make it harder for users to give each warning the same level of attention. In clinical decision-support systems, users became less likely to respond to alerts as the number of alerts and repeated reminders increased, suggesting that frequent prompts can make important information harder to distinguish from routine notifications [2]. However, this effect does not always result from simple desensitization alone, showing that factors such as cognitive load and repetition can work together [2]. Similar effects have been observed with computer security warnings, where repeated exposure was linked to decreasing responses to the warnings and greater signs of mind-wandering over time [5]. Combined with evidence that attention plays an important role in monitoring automated systems [7], these findings suggest that repeated security prompts may gradually receive less careful attention.

Research on Human Oversight in AI and Agentic Systems -

Human oversight is only useful when users stay actively involved in evaluating what an AI system is doing. Good human–AI interaction should help users understand what the system can do, recognise when it may be wrong, and correct or stop its actions when needed [1]. However, simply keeping a human in the loop does not always prevent overreliance. Users may still accept incorrect AI recommendations, even when the system provides explanations [3]. Studies suggest that people make better judgments when the interface encourages them to think about the AI's recommendation before accepting it [3]. This suggests that effective oversight depends not only on giving users control, but also on keeping them attentive and engaged while making decisions.

Research on Confirmation Frequency in AI-Agent Systems -

Recent work on AI agents shows that confirmation timing can affect both user control and the efficiency of completing a task. Asking users to approve every step gives them more opportunities to intervene, but can also create unnecessary interruptions, while waiting until the end may allow mistakes to continue through several actions [10]. Intermediate confirmation points have been found to reduce task completion time and were preferred by most users compared with confirming only at the end [10]. Permission research also shows that deciding when an AI agent should act automatically and when it should ask the user is becoming an important part of agent design [9]. However, these studies mainly focus on task efficiency, permission preferences, and reducing unnecessary interaction, rather than whether frequent confirmations change a user's ability to notice and reject an unsafe action.

Research Gap:

10 Existing research shows that people can rely too heavily on automated systems, especially when tasks require sustained attention or extra effort to verify decisions [4], [6]–[8]. Repeated alerts and security warnings can also receive less attention over time [2], [5]. In AI systems, meaningful oversight depends on users staying actively involved rather than simply approving or rejecting actions [1], [3]. Recent studies have started exploring when AI agents should ask for confirmation and how permissions can be managed efficiently [9], [10]. However, there is still limited research on whether the frequency of these requests affects users' ability to recognise and reject unsafe actions, as well as their decision speed and confidence.

Methodology:

This study will use a controlled between-subjects experiment to test whether confirmation frequency affects how carefully users judge an AI agent's actions. Around 60 participants aged 16–21 will be randomly divided into three equal groups with low, moderate, and high confirmation-frequency conditions. A target of 60 participants was chosen to provide 20 participants per condition, allowing a balanced comparison while remaining feasible for the scope of the study.

Participants will use a simulated AI-agent interface carrying out realistic digital actions, such as organising files, sharing information, or changing settings. In the first phase, everyone will see the same 15 routine actions, but the number of confirmation requests will vary: 5, 10, or 15 depending on the group. Actions without a confirmation request will appear to continue automatically.

Next, all participants will complete the same security task with 10 new actions: 5 safe and 5 unsafe. Unsafe actions will be predefined as actions that could create a clear privacy or security risk. For each action, participants will choose Allow or Reject and rate their confidence from 1 (not confident) to 5 (very confident). The actions will be shown in a random order. After completing the security task, participants will rate how repetitive and mentally tiring they found the confirmation requests presented during the first phase on a 1–5 scale. This will act as a manipulation check to see whether higher confirmation frequencies were experienced as more fatiguing.

The primary measure will be unsafe-action accuracy, scored out of 5. Overall accuracy across all 10 actions will also be recorded. Decision time will be measured as the average time taken per action, while confidence will be calculated using the average 1–5 rating. The scenarios will be pilot-tested beforehand to ensure they are clear and balanced.

The three groups will be compared for each outcome using a one-way ANOVA where its assumptions are met; otherwise, a Kruskal–Wallis test will be used. Relevant effect sizes will also be reported.

Project Practicalities -

The study will involve around 60 participants, with roughly 20 participants in each condition. Each session is expected to take about 15–20 minutes and can be completed using a simple browser-based prototype on a laptop or computer.

The project will be carried out in stages: designing and testing the scenarios, running a small pilot study, recruiting participants, conducting the experiment, and finally analysing the results. Participants will be informed about the study beforehand and participation will be voluntary. Responses will be recorded anonymously, and participants will be free to leave the study at any time. Since the experiment uses only a simulated AI-agent interface, no real accounts, files, permissions, or personal data will be accessed or changed. For participants under 18, parental or guardian consent will be obtained where required.

Potential Limitations -

A limitation is that a simulated AI agent interface won't fully reflect how people behave when real files, accounts, or data are at risk. The short experiment will mainly capture short-term confirmation fatigue, whereas real users might interact with AI agents over a much longer period.

The participant group may mostly consist of students, so the results may not represent all types of users. Differences in AI experience could also affect how carefully participants judge the agent's actions. To reduce these effects, participants will be randomly assigned to groups,

everyone will receive the same final security task, and the scenarios will be pilot-tested beforehand to make sure they are clear and reasonably balanced.

Conclusion -

This method will clearly examine how different confirmation frequencies influence users' decisions. The study can demonstrate whether frequent confirmations lead to less careful oversight by measuring accuracy, decision time, and confidence, and thus help identify the appropriate balance between human control and AI-agent autonomy.

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